**a Coverage and stage of historical Galaxy request failures**

| Benchmark | Traced runs | Before a job ran | During execution | Other | All failures |
|---|---|---|---|---|---|
| BixBench | 600 | 895 | 369 | 78 | 1,342 |
| CompBioBench | 1,188 | 2,052 | 2,650 | 678 | 5,380 |
| IWC | 108 | 150 | 162 | 64 | 376 |

**b Failed requests by class and benchmark**

Counts are printed in each cell. Shading shows the share of failures within a benchmark.

0% >0–<5% 5–<10% 10–<20% ≥20%

| Stage | Failure class | BixBench | CompBioBench | IWC |
|---|---|---|---|---|
| **Before job** | Tool schema needs history context | 483 | 590 | 48 |
| | Tool identifier missing or guessed | 49 | 316 | 19 |
| | Nested parameter structure | 79 | 123 | 14 |
| | Parameter value or datatype | 150 | 566 | 49 |
| | Invalid or foreign identifier | 98 | 132 | 12 |
| | User-defined tool schema | 9 | 117 | 0 |
| | Upload or datatype registry | 3 | 91 | 0 |
| | Server, transport or rate limit | 24 | 117 | 8 |
| **During job** | Missing container dependency | 12 | 68 | 0 |
| | Job error without diagnostic | 24 | 1,242 | 73 |
| | Tool runtime error | 317 | 1,071 | 70 |
| | Input format, compression or index | 16 | 176 | 17 |
| | Memory or resource limit | 0 | 93 | 2 |
| **Other** | Additional exceptions | 75 | 458 | 35 |
| | Unclassified | 3 | 220 | 29 |